IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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9/12/2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies

1) Data collection – web scraping and api calls using static URLs
2) Data wrangling – processed null values to ensure relevant analysis outcomes, identified mission outcomes (success vs failures) by regions both land and water-based
3) Exploratory data analysis (EDA) - used visualizations and SQL including matplotlib and seaborn to analyze rocket weight vs landing success
4) Interactive visual analytics - using Folium and Plotly Dash to present findings in map form to better understand data relationships and browser dashboards for custom reporting
5) Predictive modeling - built, evaluated and deployed predictive models using scikit-learn tools including linear regression, decision tree and Kneighbors classifiers to select the best predictive model for successful launch outcomes
6) Insights – An accurate predictive Falcon 9 launch AI model was developed and is currently available for use.

Executive Summary

- Summary of all results

1) Launch Trends – the rate of successful launches improved consistently over time across a range from 2010-2020 , showing growing reliability for rocket recoveries. The only exception being the year 2018 which saw a decline

2) Launch site impact and payload weight – CCAFS SLC 40 specialized in heavier payloads with success, KSC LC 39A was most reliable with lighter payloads and VAFB SLC 4E had the highest success rate although with substantially fewer launches and lighter payloads compared to other sites

3) Payload impact considering the orbit type – heavier payloads were most successful for PO, ISS and VLEO orbits while SSO had the highest launch and landing success rate among lighter payloads

4) Conclusions - Successful landings depend largely on the launch site, payload weight and type of orbit. Reuseable rocket technology for SpaceX is reliable, repeatable and a competitive advantage saving over $100M per launch

Introduction

- Project background and context
  - In this project, the overall goal was to build a model that would predict if the Falcon 9 first stage booster rocket would land successfully. If successful, the landing of the booster saves about $100 Million per launch by being able to reuse the stage 1 rocket giving SpaceX a significant advantage over its competition
- Problems you want to find answers
  - Can the Falcon 9 stage 1 boosters land safely? If yes, then we can calculate the cost of future launches compared to the competition. How often does the booster land successfully? These percentages can then be applied to properly weigh the cost calculations

Section 1

Methodology



Methodology

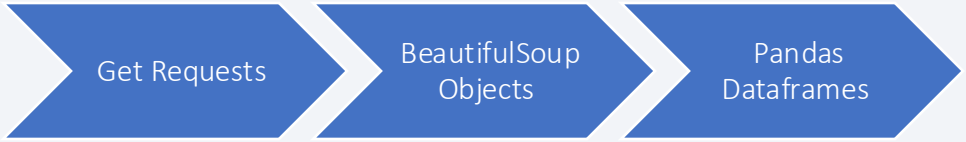
Executive Summary

- Data collection methodology:
  - Web-based JSON GET requests, data parsing using json.normalize(), BeautifulSoup objects to parse HTML content
- Perform data wrangling
  - Data was processed using pandas to import data into a dataframe for analysis
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Used scikit-learn for train_test_split, DecisionTree and KNeighbors classifiers among many others

Data Collection

- How data sets were collected
  - Referenced SpaceX API endpoints using requests.get() and then converted the response content into pandas dataframes using the pandas library and the json.normalize() method to efficiently extract relevant data
  - Additionally tabular data was scraped from static web URLs using BeautifulSoup objects
- Data Collection process flowcharts



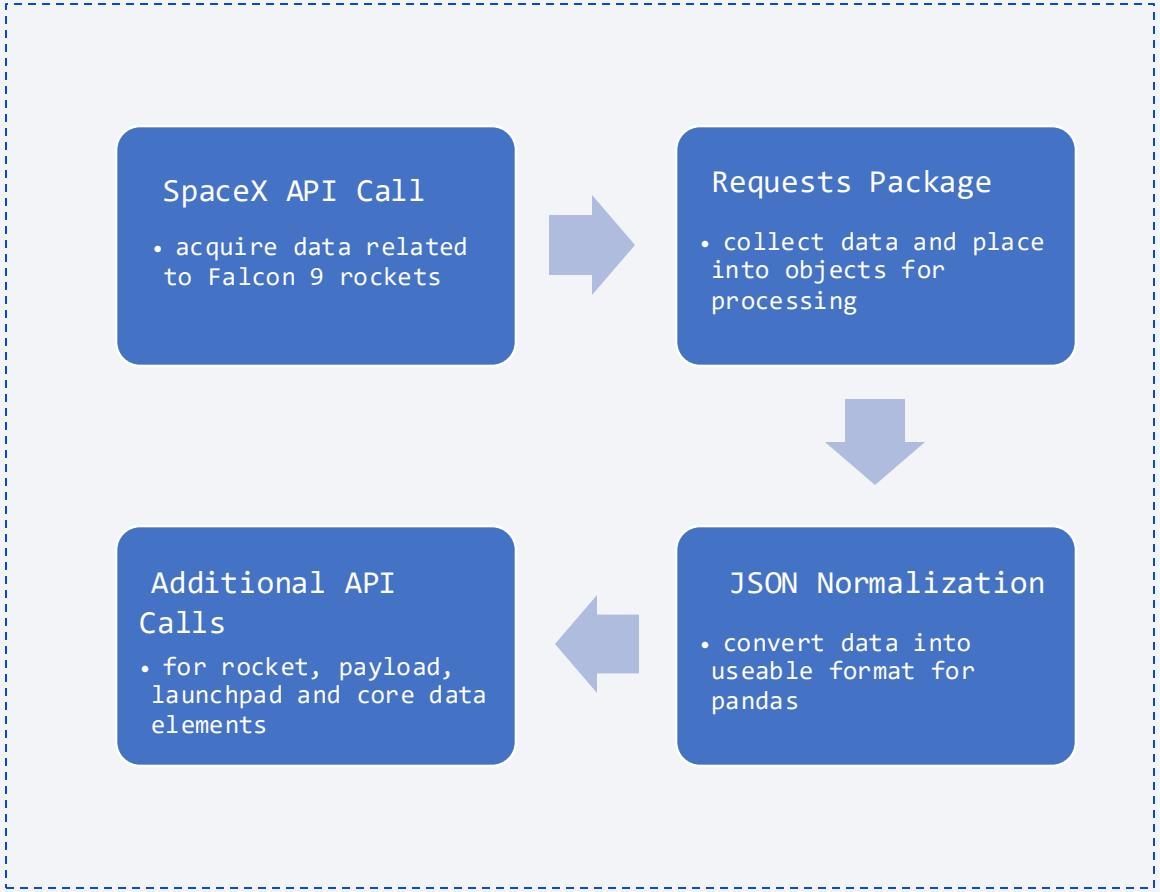


Data Collection – SpaceX API

- API Functions – used to help extract information using ID numbers in the launch data
- Primary API call to the root SpaceX endpoint
- Initial Data Print – to validate response content
- Parse and Normalize Data – for efficient processing using the pandas library
- Additional API calls – used to obtain more granular data relating to rocket details
- Global Variable Lists – to hold collected data and run functions against
- Convert to Dataframes – process lists into dictionaries that can be converted into a pandas dataframe for powerful analysis

GitHub URL of the completed SpaceX API calls notebook:

- https://github.com/alphoDS100/capstone-rocket/blob/main/notebooks/01_spacex-data-collection-api-v2.ipynb

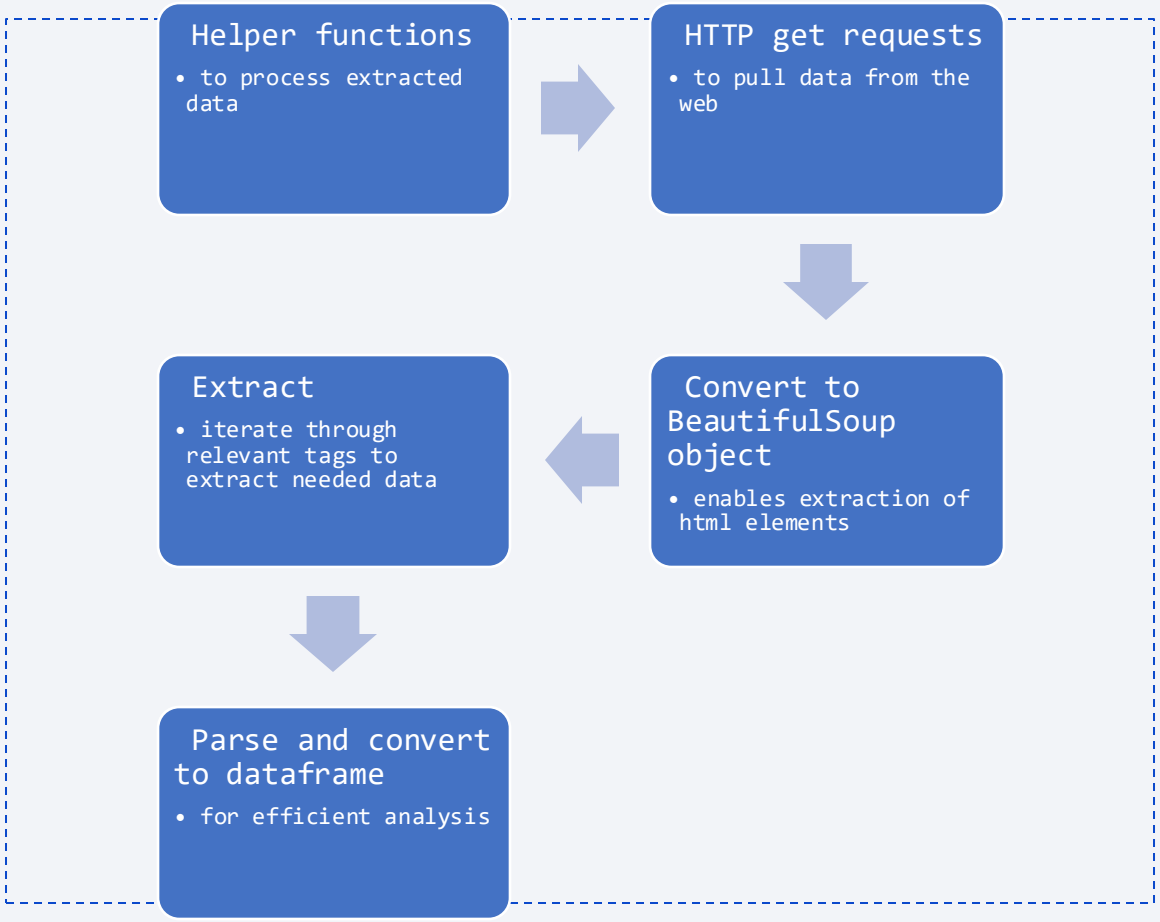


Data Collection - Scraping

- Extract – gather launch records using BeautifulSoup from online HTML table
- Parse the data – convert into useful format for further processing
- Format conversion – transform data into pandas dataframe to run advanced analysis

GitHub URL of the completed web scraping notebook:

- https://github.com/alphoDS100/capstone-rocket/blob/main/notebooks/02_spacex-webscraping.ipynb

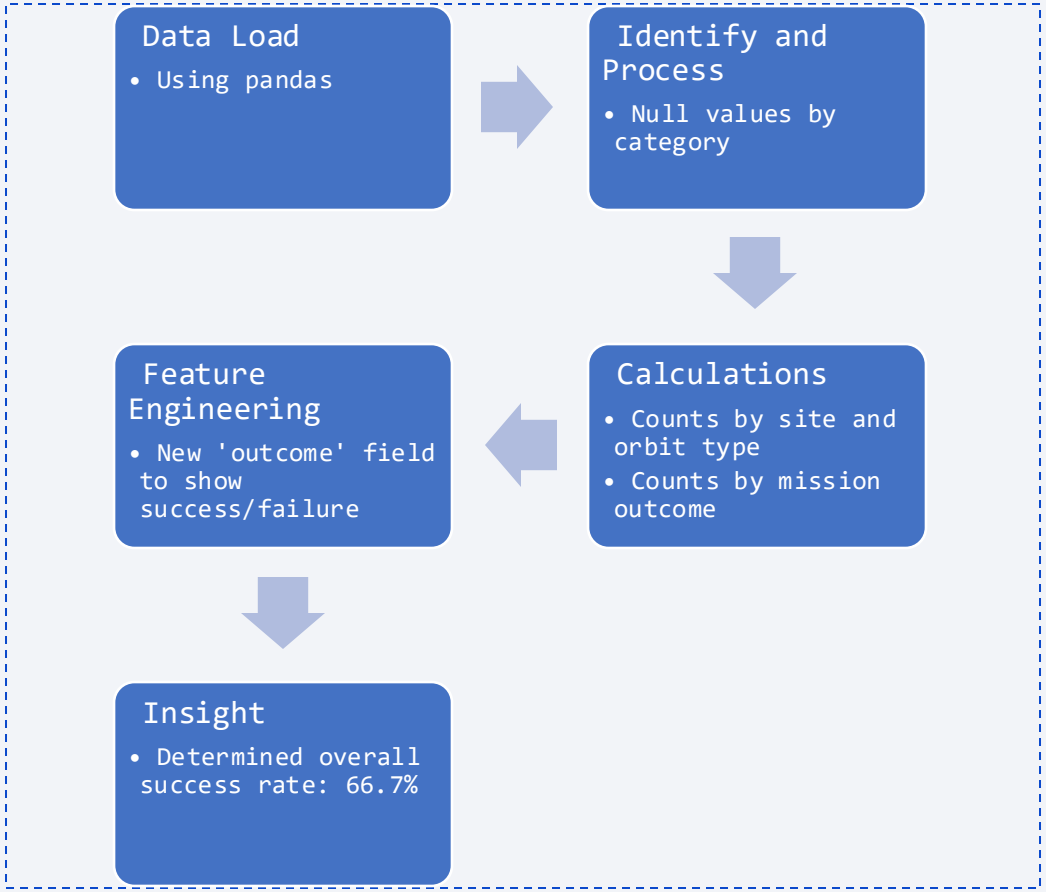


Data Wrangling

- Data import – using pandas library
- Missing values and data types – identification to aid in further analysis
- Categorical counts – how many launch sites, successes, failures, orbits, rockets, etc.
- Conversion – categorical values into binary for easier calculations

GitHub URL of completed data:

- https://github.com/alphoDS100/capstone-rocket/blob/main/notebooks/03_spacex-Data%20wrangling-v2.ipynb



EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts
  - Chart Type 1: Seaborn catplot – this plot is great for showing a comparison between numeric and categorical variables, in this case, a chart was built for continuous flight numbers, a numeric payload mass and a categorical successful/unsuccessful landings attribute
  - Chart Type 2: Seaborn scatterplot – used to compare a categorical launch site field to a numeric payload mass and binary field of 'class' which translates to a failed or successful launch
  - Chart Type 3: Bar plot – used to best represent and compare the success and failure percentage by categorical variable 'orbit'
  - Chart Type 4: Line graph – selected to showcase a binary result of launch success/failure over the continuous variable of time

GitHub URL of the completed EDA notebook:

- https://github.com/alphoDS100/capstone-rocket/blob/main/notebooks/04_eda-dataviz-v2.ipynb

EDA with SQL

Summarize the SQL queries you performed

- ```
  create table SPACEXTABLE as select * from SPACEXTBL where Date is not null
  ```
- ```
  SELECT * FROM SPACEXTBL LIMIT 3
  ```
- ```
  SELECT DISTINCT "Launch_Site" FROM SPACEXTBL
  ```
- ```
  SELECT * FROM SPACEXTBL WHERE "Launch_Site" like 'CCA%' LIMIT 5
  ```
- ```
  SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD
      FROM SPACEXTBL
      WHERE Customer = "NASA (CRS)"
  ```
- ```
  SELECT Booster_Version
  FROM SPACEXTBL
  WHERE (Landing_Outcome = 'Success (drone ship)')
  AND (PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000)
  ```
- ```
  SELECT Landing_Outcome, COUNT(*) AS "Landing_counts" FROM SPACEXTBL
      WHERE (Date BETWEEN '2010-06-04' AND '2017-03-20')
      GROUP BY Landing_Outcome ORDER BY Landing_counts DESC
  ```

GitHub URL of the completed EDA with SQL notebook:

- https://github.com/alphoDS100/capstone-rocket/blob/main/notebooks/05_eda-sql-coursera_sqllite.ipynb
  ```

Build an Interactive Map with Folium

Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map

- Circle – used to call attention to areas of interest and further compliment latitude and longitude coordinates. Circles are a good complement to markers
- Marker – can provide additional detailed information about a particular point on a map and were used to specifically show locations of launch sites in Florida and California
- MarkerClusters – are a good way to simplify a map containing many markers that share the same coordinates. The folium maps show clusters relating to successful and failed launch campaigns across the various launch sites
- PolyLine – used to show distance in between two areas of interest, e.g. launch sites and closest railways, cities, roads and coastlines

GitHub URL of the completed interactive map with Folium:

- https://github.com/alphoDS100/capstone-rocket/blob/main/notebooks/06_folium_launch-site-location-v2.ipynb

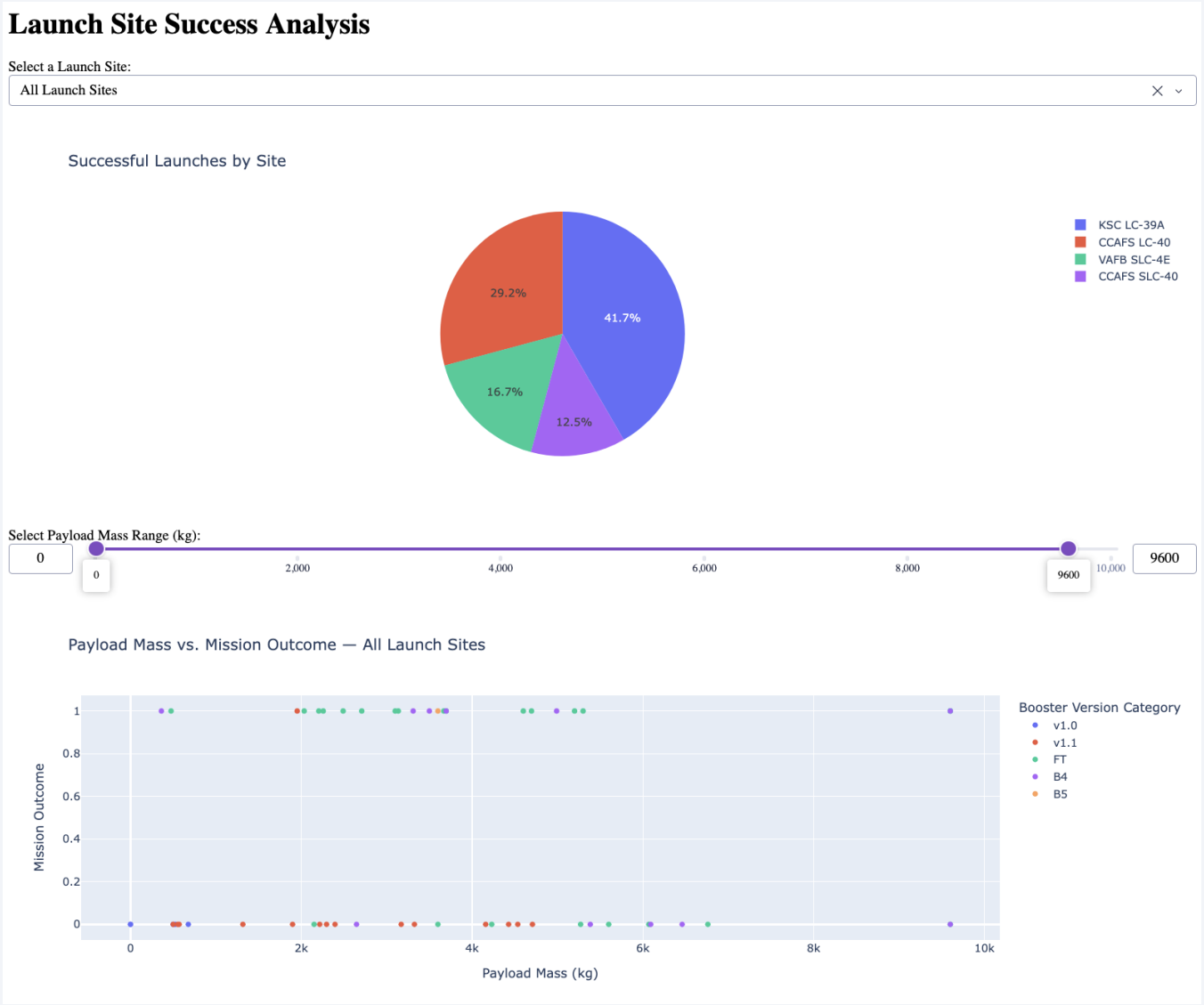
Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- The dashboard starts at the top by inquiring of the user what data they would like to see populate the included graphics. Dropdowns were added to let a user select appropriate site locations or all. Also, a slider selector was built to allow for ad-hoc user input in selecting a payload mass range
- A pie chart was used to show relative percentages across all launch sites comparing successful launches. For individual sites, a pie chart compare successful launches against failed launches
- In addition, a scatter plot was integrated to display payload mass information, mission outcomes, booster versions and launch site content in a single plot for easy attribute comparison across 4 primary features

GitHub URL of the completed Plotly Dash lab:

- https://github.com/alphoDS100/capstone-rocket/blob/main/notebooks/07_finalfinal_dash_notebook.ipynb

Build a Dashboard with Plotly Dash 2



Predictive Analysis (Classification)

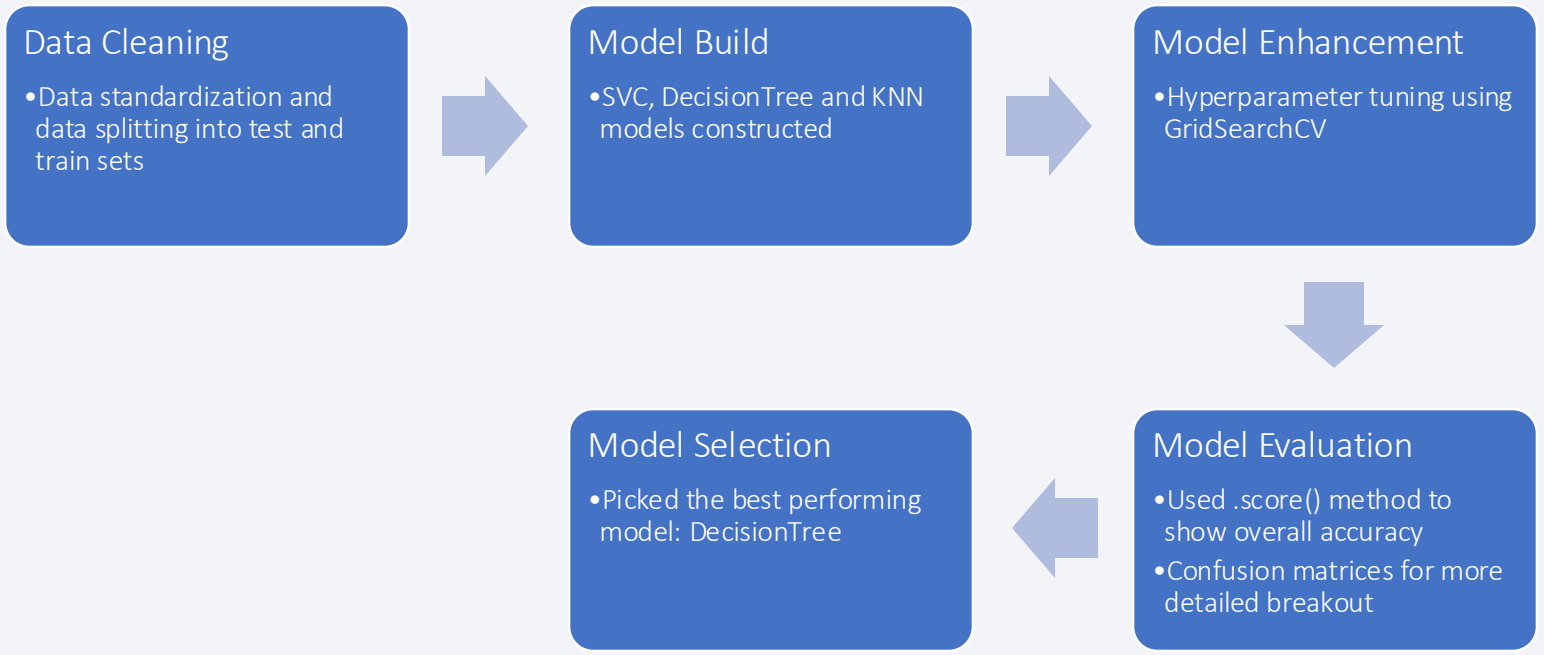
Summarize how you built, evaluated, improved, and found the best performing classification model

- A predictive model was built to determine if a first stage SpaceX rocket would land successfully. By determining the outcome, this information can be used to further enhance and support SpaceX's competitive advantage in a multi-bid situation
- Data cleaning - performed first to ensure that data could be used efficiently for model building. The data was then standardized to enhance model performance. Following, data was split into test and train subsets
- Multiple models built - pipelines were structured to test multiple ML classifiers including: SVM, Classification Trees and Logistic Regression. Hyperparameter evaluation was executed and values modified to build the best performing model in each category. GridSearch Cross Validation was also deployed to help balance model outcomes
- Model evaluation - used data parameters: best_params_ and best_score_. In addition, a confusion matrix was built to calculate model performance and identify false positives and negatives
- Model Selection – classification tree models, namely, the DecisionTreeClassifier in scikit-learn performed the best overall with the highest accuracy and lowest number of false positives and false negatives

GitHub URL of the completed predictive analysis lab:

- https://github.com/alphoDS100/capstone-rocket/blob/main/notebooks/08_SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Predictive Analysis (Classification) Flow Chart



Results

Model Comparisons

LogisticRegression

Accuracy using Training Data: 0.8464

Accuracy using Test Data: 0.83

SVC (Support Vector)

Accuracy using Training Data: 0.8482

Accuracy using Test Data: 0.83

DecisionTree

Accuracy using Training Data: 0.8625

Accuracy using Test Data: 0.83

KNeighbors

Accuracy using Training Data: 0.8482

Accuracy using Test Data: 0.83

The Decision Tree model performed best with the highest cross-validation score of 86.25% with all models showing a similar test accuracy score of 83%.

EDA Findings

Missing Values: LandingPad had 28.89% missing values, while all other columns were fully populated.

Data Type Identification: There were 4 numeric attributes, 9 categorical attributes and 4 continuous attributes

Total Launches

LaunchSite	
CCAFS SLC 40	55
KSC LC 39A	22
VAFB SLC 4E	13

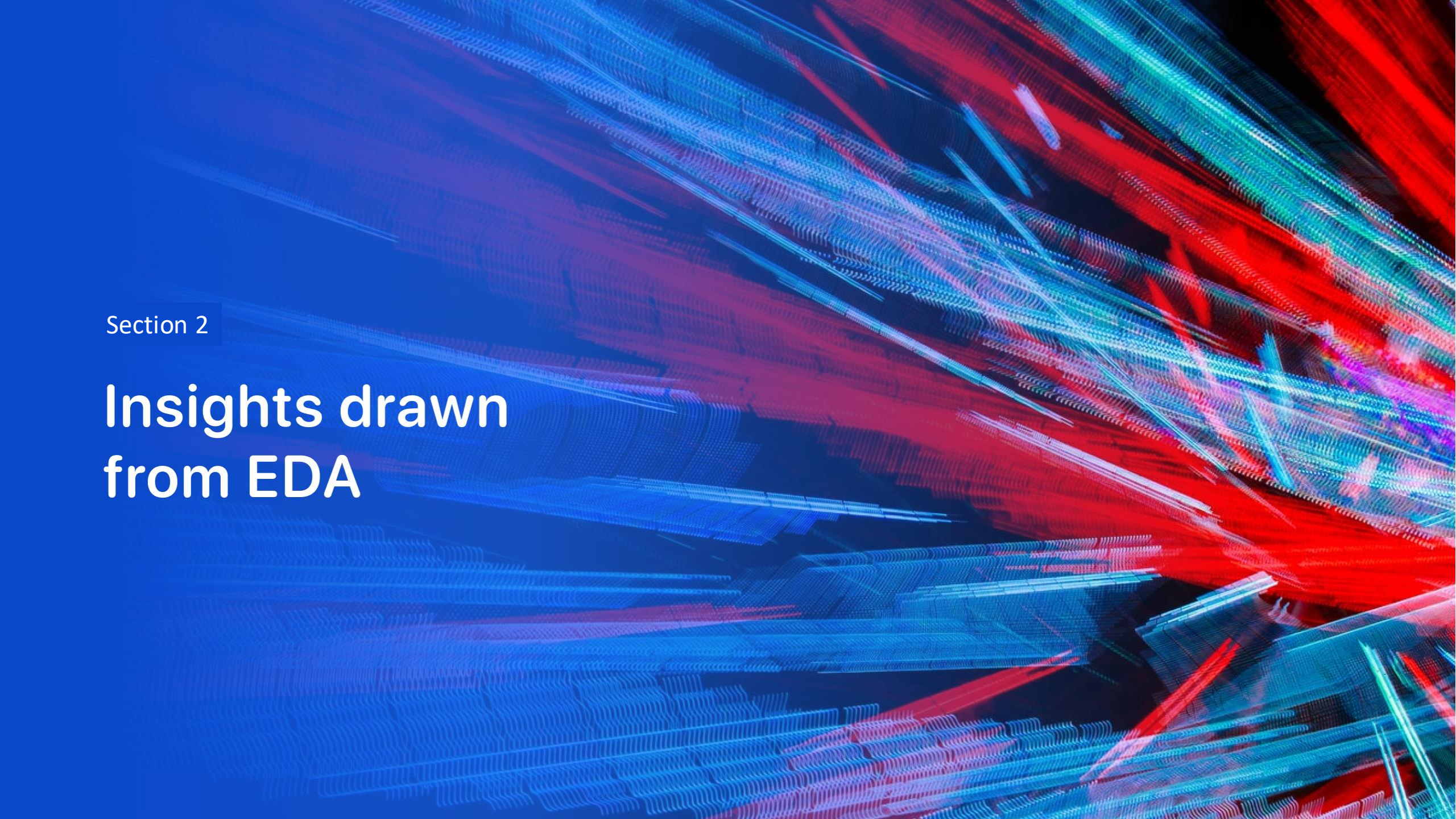
Orbit Counts by Type

ISS	21
VLEO	14
PO	9
LEO	7
SSO	5
MEO	3
ES-L1	1
HEO	1
SO	1
GEO	1

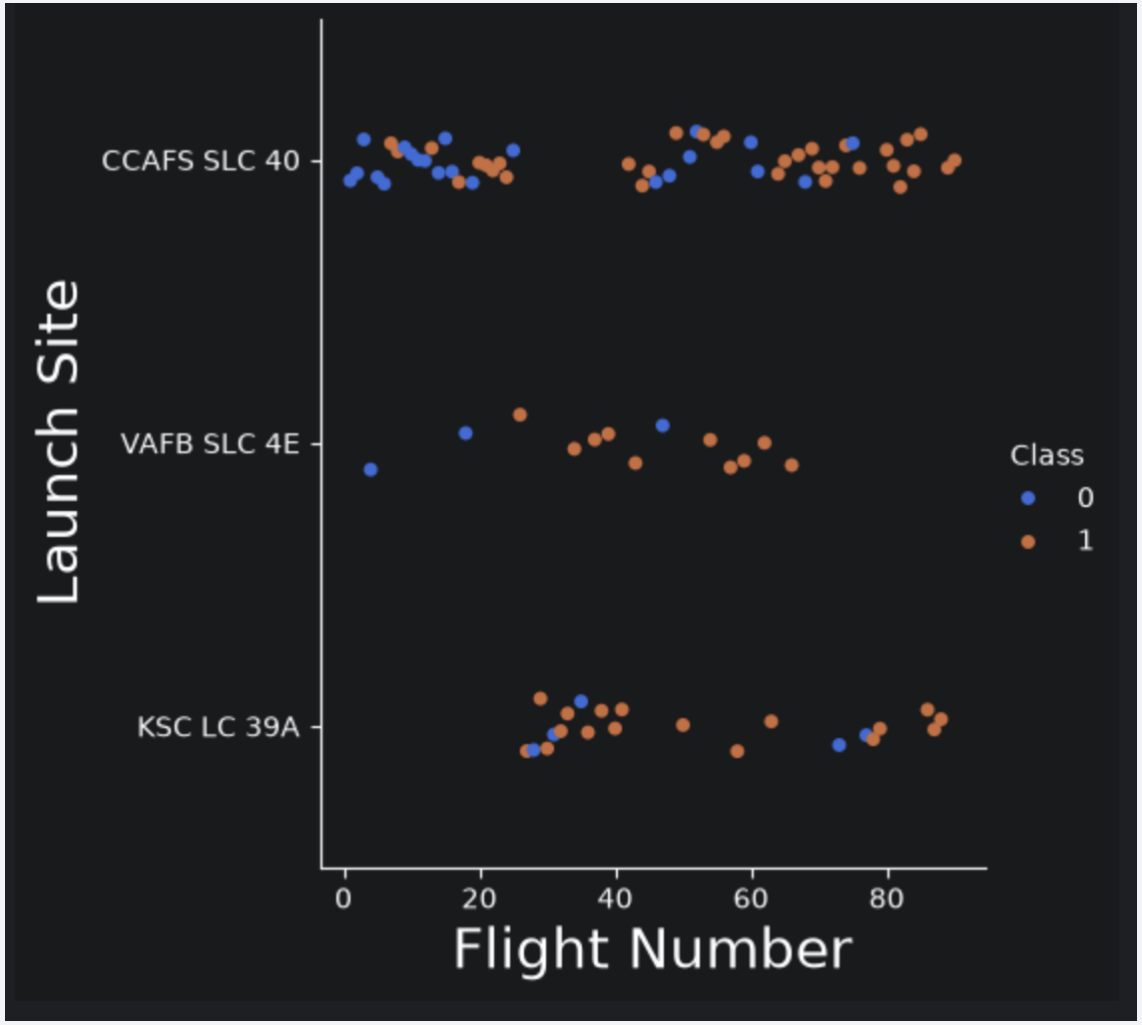
Outcomes by Landing Type

Outcome	
True ASDS	41
None None	19
True RTLS	14
False ASDS	6
True Ocean	5
False Ocean	2
None ASDS	2
False RTLS	1

Stage 1 Rocket Recovery Success Rate: 66.67%


Section 2
Insights drawn from EDA

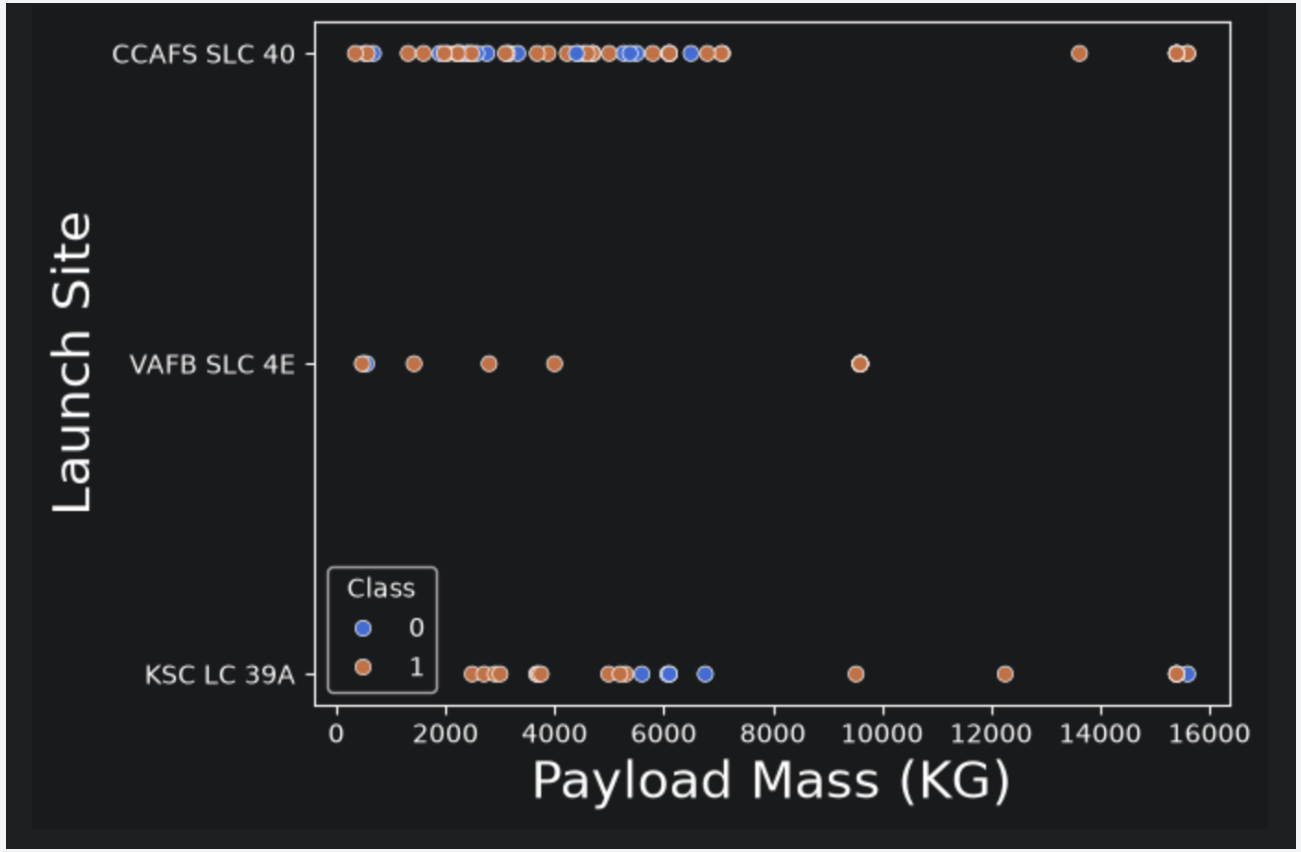
Flight Number vs. Launch Site



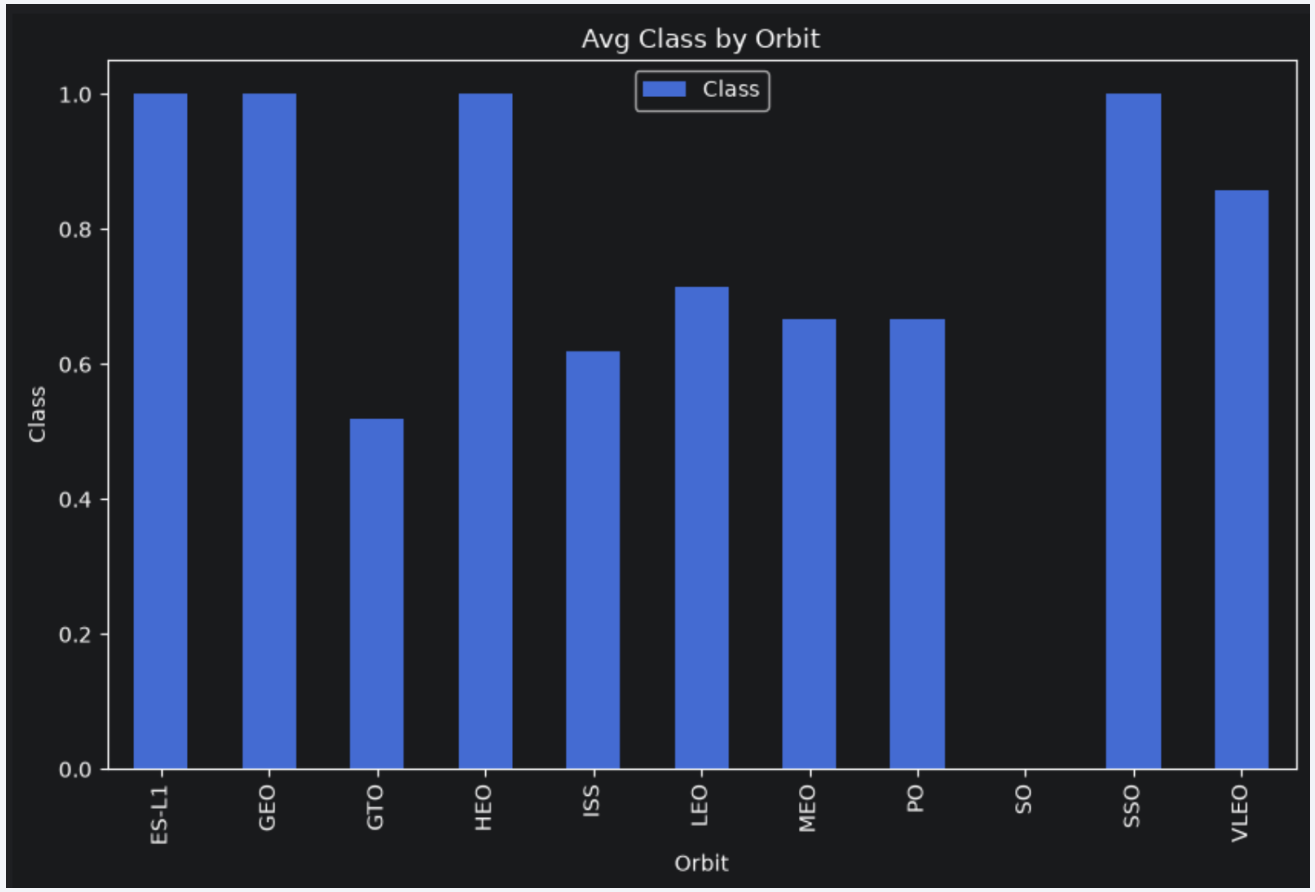
- CCAFS SLC 40: Exhibits the most launches overall. Early flights were regularly unsuccessful. There is consistent improvement over time with later flights having more success in recovery of stage 1 rockets
- VAFB SLC 4E: The lowest number of launches total of any site. A couple of early failed recoveries with high performance following thereafter
- KSC LC 39A: The best performing site by ratio. This could be a result of starting later in the flight number launch cycle with site operators and engineers learning from other launch site location failures and improving upon previous flight plans
- General observations: More flight attempts lead to more successful recoveries. As flight tests are carried out it is clear that the teams learn from failed outcomes and improve processes as well as capture technologies.

Payload vs. Launch Site

- CCAFS SLC 40: Exhibits the most launches overall. Shows a wide range of payload variance which is weighted more heavily toward lighter mass missions. Regardless, heavy payload flights were consistently successful
- VAFB SLC 4E: The lowest number of launch payload of any site. This location primarily tested low weight missions with more successful missions than failed missions
- KSC LC 39A: Covered the entire spectrum of payload options. Tended to be better suited to lighter payloads, weakest in the mid-range and better than 50% success for heavy payloads
- General observations: There is a decent balance of testing occurring across the 3 sites. CCAFS SLC 40 supported light and heavy payloads, with VAFB SLC 4E focused more on light payloads, KSC LC39A covered the entire range of payload mission weights



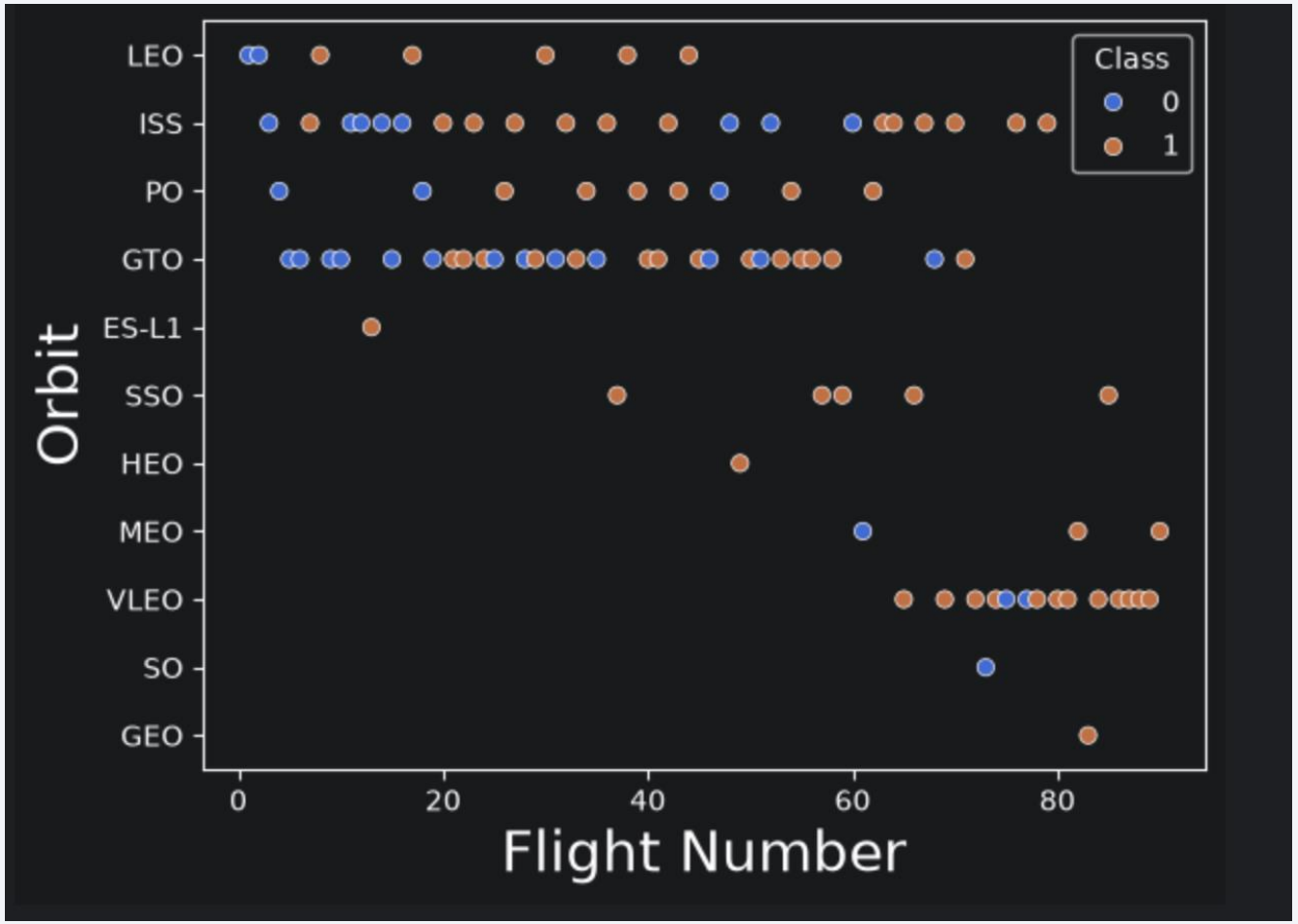
Success Rate vs. Orbit Type



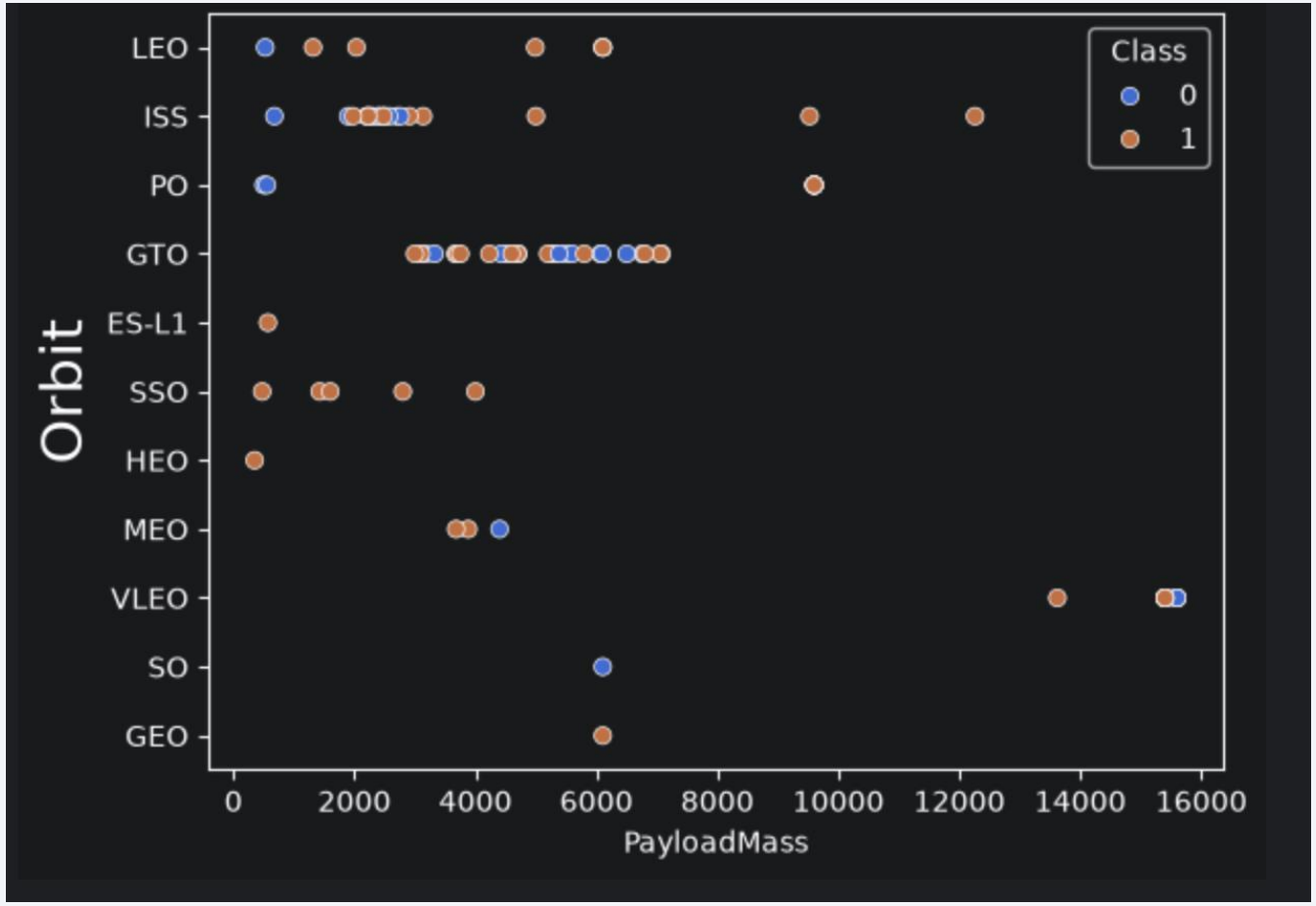
- General observations: ES-L1, GEO, HEO, SSO orbit types scored the highest in terms of successful recoveries. All others fell into a middle ground for performance with SO having the lowest score of 0.0

Flight Number vs. Orbit Type

- General observations: The LEO orbit appears to have better success with more flights being executed. This is not the case for GTO that has intermittent failures through its flight number missions. Several orbit types likely benefited from the knowledge gained in earlier missions with success being much higher for later flight numbers as seen in SSO and VLEO



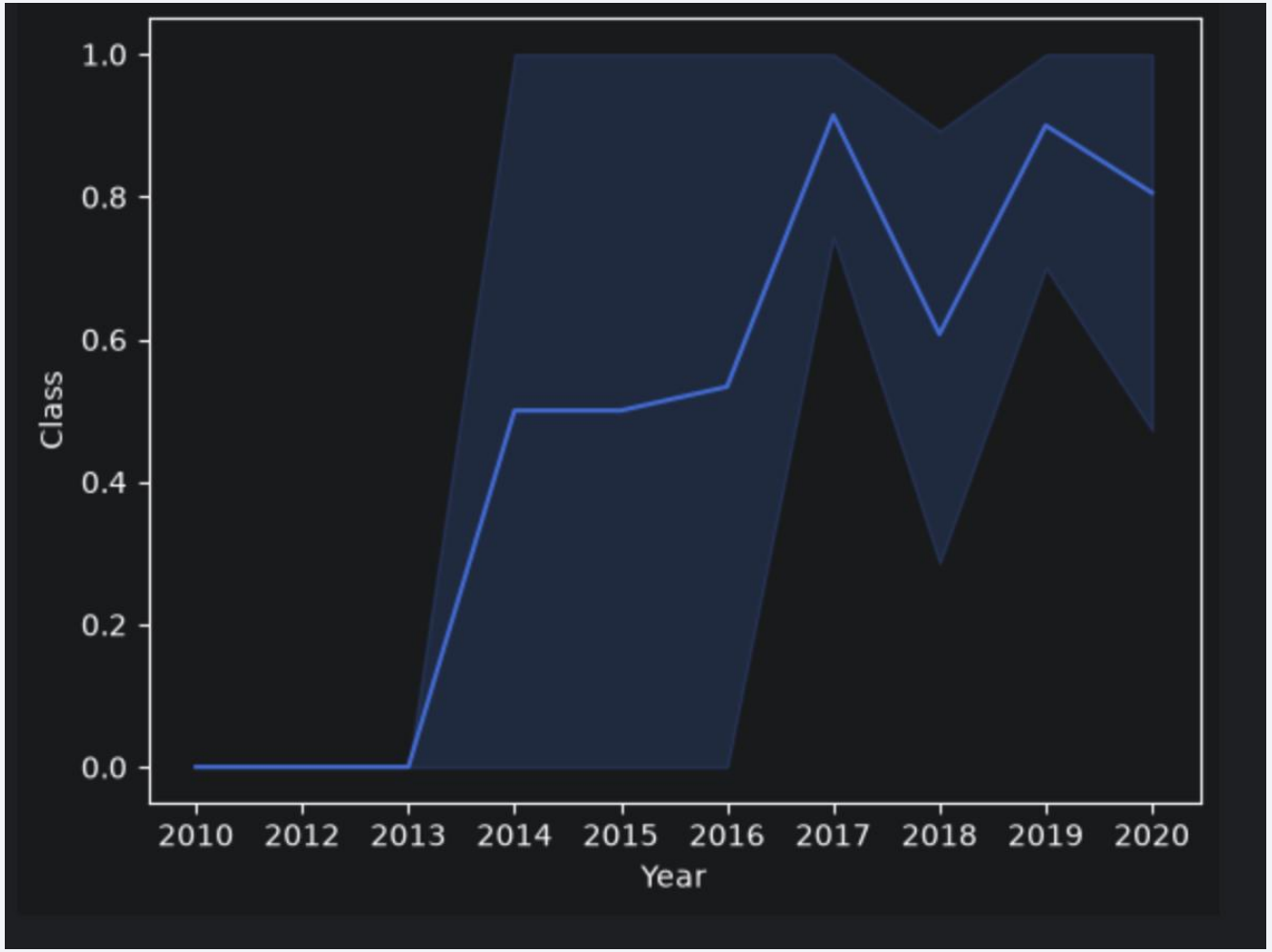
Payload vs. Orbit Type



- General observations: For heavier payloads the positive landing rates are higher for ISS, PO and VLEO
- The same cannot be said for GTO as there is a balanced mix of successful and failed rocket recovery missions

Launch Success Yearly Trend

- General observations: The success rate since 2013 continued to increase until 2017 when there was a slight decrease into 2018. Things quickly picked back up climbing above 80% success into 2020



All Launch Site Names

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTBL
[9]
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Used SQL to quickly query the distinct launch site values found in the SpaceX table

Launch Site Names Begin with 'CCA'

- A SQL SELECT operation was used in combination with a WHERE filter to target the relevant field holding the site names. A LIKE filter was added to narrow the search further to 'CCA'

```
%sql SELECT * FROM SPACEXTBL WHERE "Launch_Site" like 'CCA%' LIMIT 5
[10]
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)

Total Payload Mass

```
%%sql
SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD
    FROM SPACEXTBL
    WHERE Customer = "NASA (CRS)"
[14]
```

```
 * sqlite:///my_data1.db
Done.
```

TOTAL_PAYLOAD
45596

- A WHERE filter was used to narrow the data to NASA. Once the data was limited, a sum filter was applied to the payload feature

Average Payload Mass by F9 v1.1

- A WHERE filter was used to narrow the search to booster_version, specifying the exact model using LIKE
- An AVG() operation was applied to the payload attribute and a new column added using AS to display the average

```
%%sql
SELECT AVG(PAYLOAD_MASS__KG_) AS AVG_PAYLOAD
    FROM SPACEXTBL
    WHERE "Booster_Version" like "F9 v1.1%"
[17]
  * sqlite:///my_data1.db
 Done.
```

AVG_PAYLOAD
2534.6666666666665

First Successful Ground Landing Date

```
%%sql
SELECT min(DATE) from SPACEXTBL
    WHERE Landing_Outcome = 'Success (ground pad)'
[23]
```

```
 * sqlite:///my_data1.db
Done.
```

min(DATE)
2015-12-22

- Found the first successful ground landing date using a WHERE filter to target the Landing_Outcome attribute with 'Success (ground pad)'
- A secondary filter was added using min(DATE) to find the earliest date

Successful Drone Ship Landing with Payload between 4000 and 6000

- A WHERE filter was used to find successful drone ship landings in combination with an AND qualifier to limit the payload to >4000 and <6000 kg
- The output was ultimately filtered to show related Booster_Version types

```
%%sql
SELECT Booster_Version
FROM SPACEXTBL
WHERE (Landing_Outcome = 'Success (drone ship)')
AND (PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000)
;
[24]
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

```
%%sql
SELECT Mission_Outcome, Count(*)
FROM SPACEXTBL
GROUP BY Mission_Outcome
;
[25]
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	Count(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- Calculated counts for successful and failure mission outcomes using the SQL Count() command as well as GROUP BY sort feature

Boosters Carried Maximum Payload

- Used a nested SELECT statement to find the Falcon 9 rocket booster versions that carried the largest payloads

```
%%sql
SELECT Booster_Version FROM SPACEXTBL
WHERE PAYLOAD_MASS__KG_ IN (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);
[26]
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

```
%%sql
SELECT Booster_Version, substr(Date, 6,2), Launch_Site
FROM SPACEXTBL
WHERE (Landing_Outcome = 'Failure (drone ship)')
AND substr(Date,0,5)='2015'
```

[30]

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	substr(Date, 6,2)	Launch_Site
F9 v1.1 B1012	01	CCAFS LC-40
F9 v1.1 B1015	04	CCAFS LC-40

- Used a substring to extract date information from the SQL table
- Then filtered output to show 'Booster_Version' and 'Launch_Site' along with associated month for 2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- A robust SQL query using many previous elements, such as, Count(), WHERE, GROUP BY, ORDER BY and DESC to organize landing outcomes between the given date range

```
%%sql
SELECT Landing_Outcome, COUNT(*) AS "Landing_counts" FROM SPACEXTBL
    WHERE (Date BETWEEN '2010-06-04' AND '2017-03-20')
    GROUP BY Landing_Outcome ORDER BY Landing_counts DESC
[32]
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	Landing_counts
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1


Section 3
Launch Sites
Proximities Analysis

Folium Map Launch Site Locations

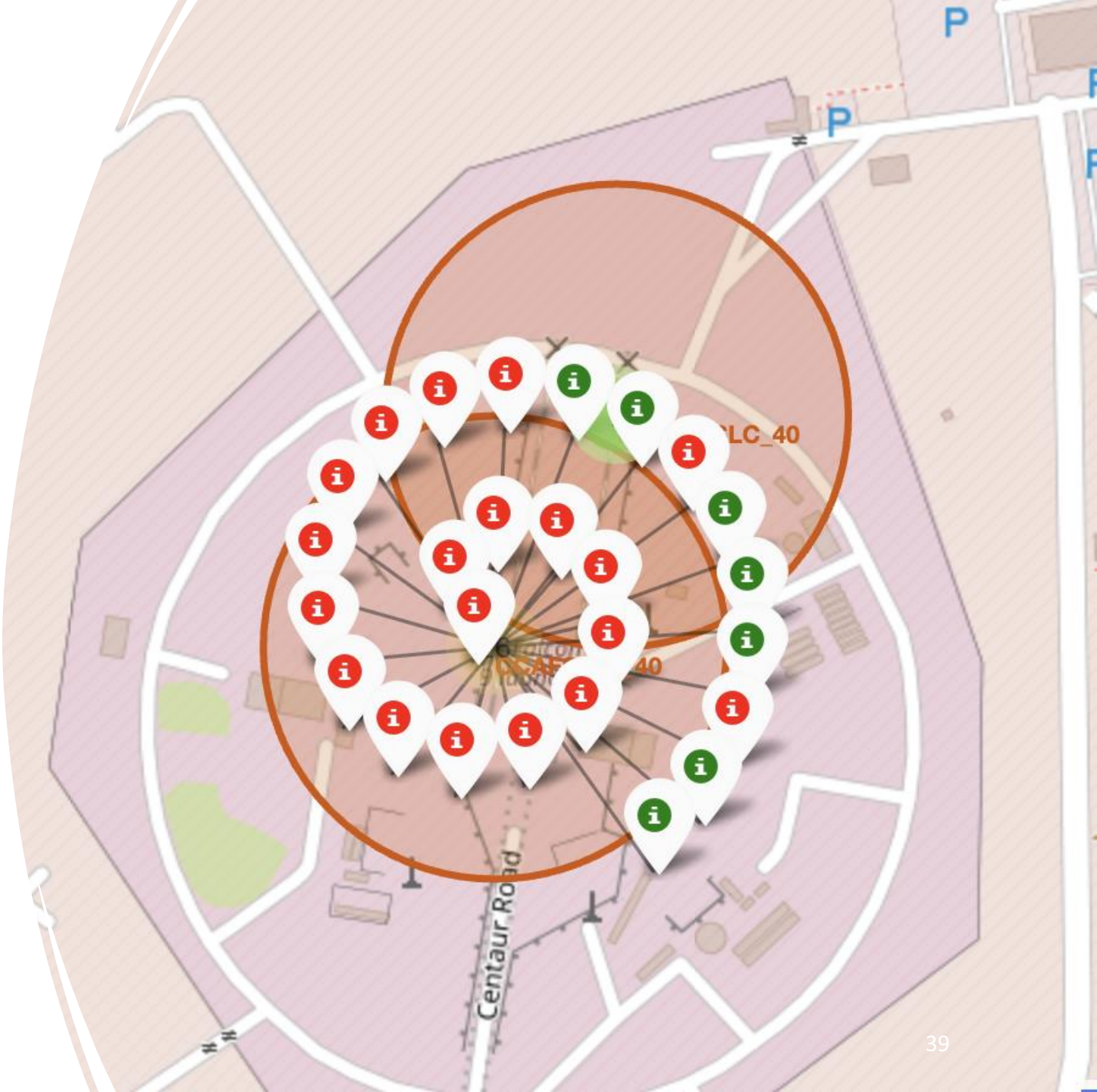
- Launch Sites located at these coordinates:



Launch Site	Lat	Long
CCAFS LC-40	28.562302	-80.577356
CCAFS SLC-40	28.563197	-80.576820
KSC LC-39A	28.573255	-80.646895
3 VAFB SLC-4E	34.632834	-120.610745

Folium Color-labeled Map

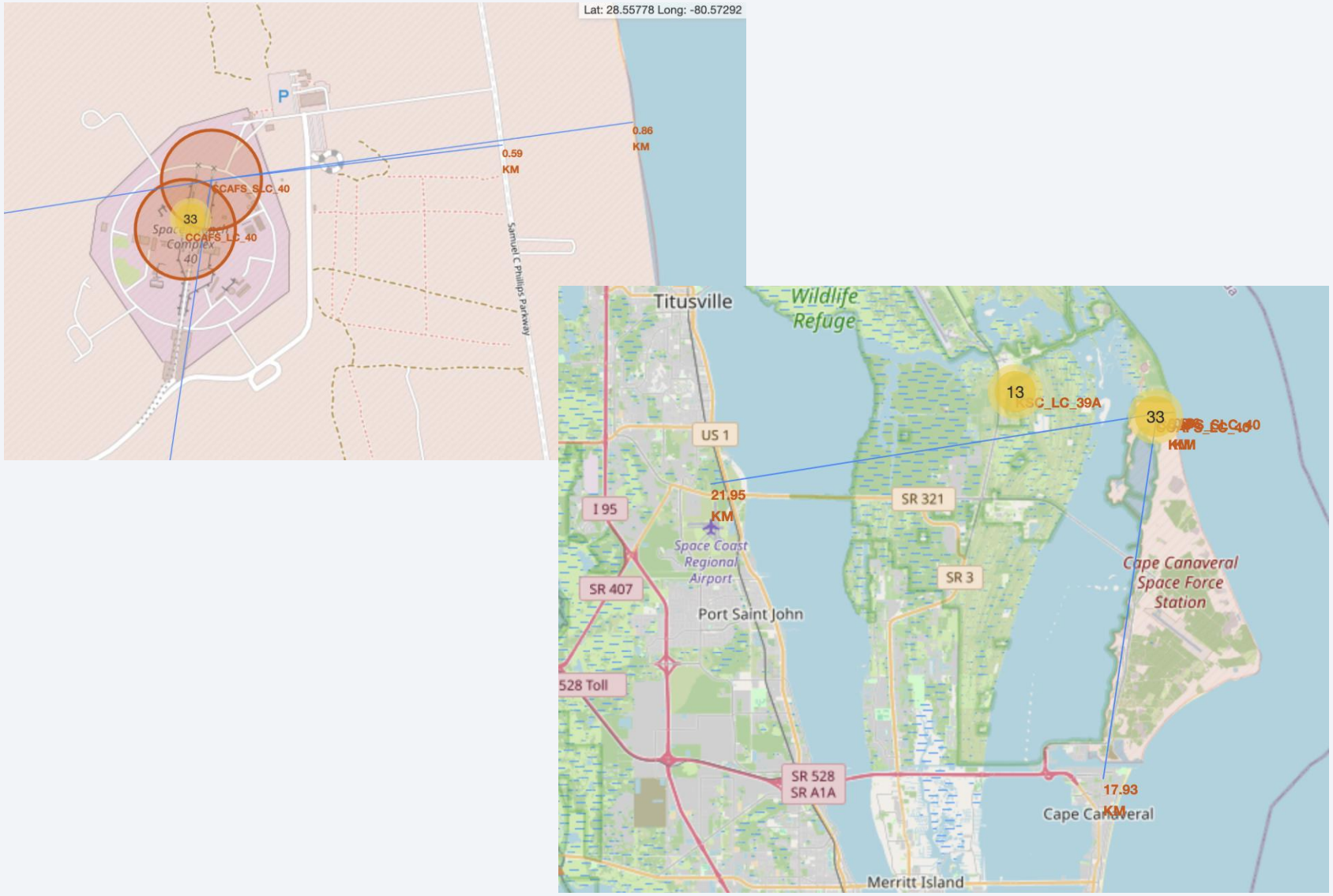
- Map showing successful (green) and failed (red) launches at site CCAFS LC 40

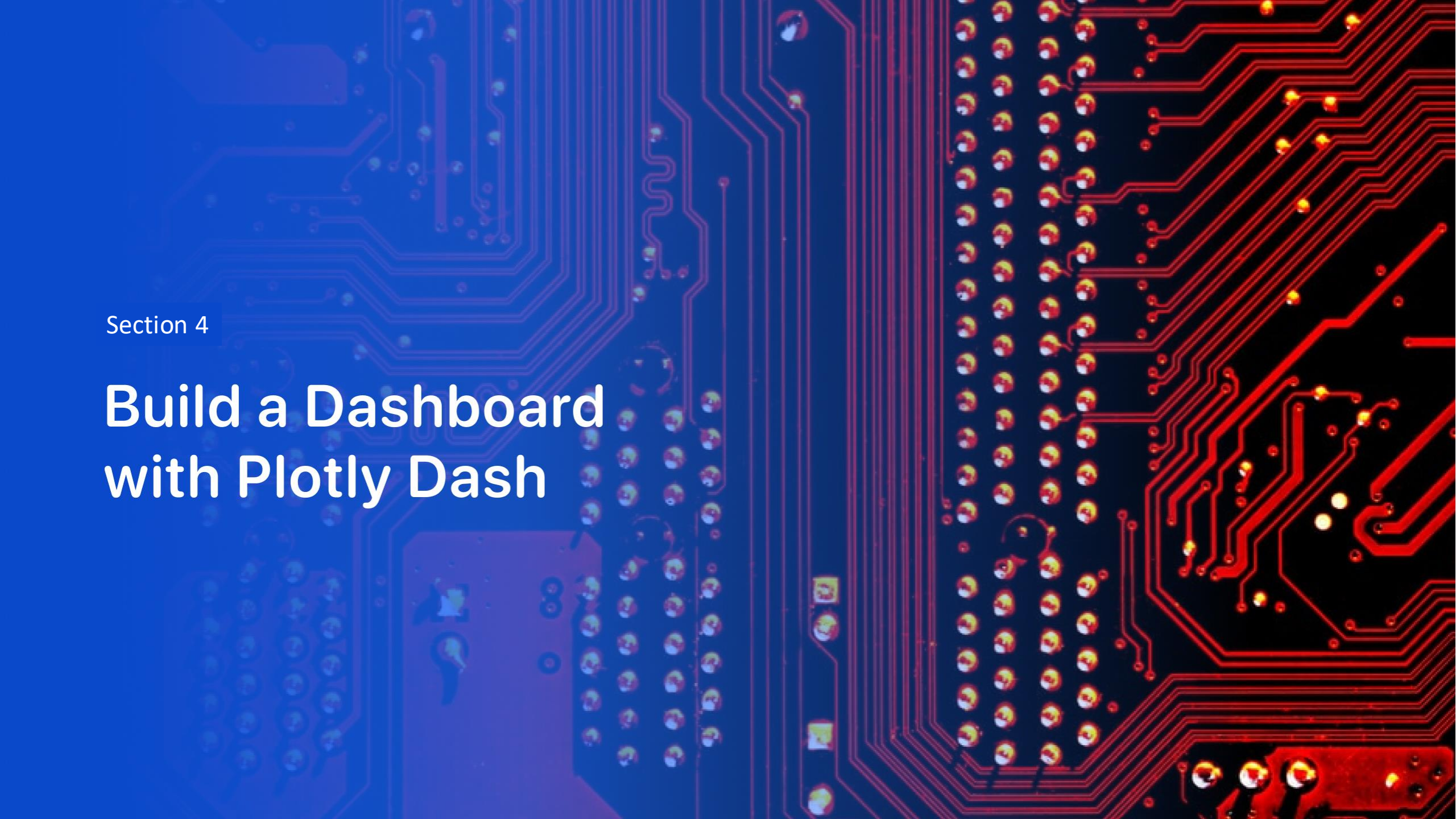


Folium Map with Relevant PolyLines

Important landmark distances from launch site CCAFS SLC 40:

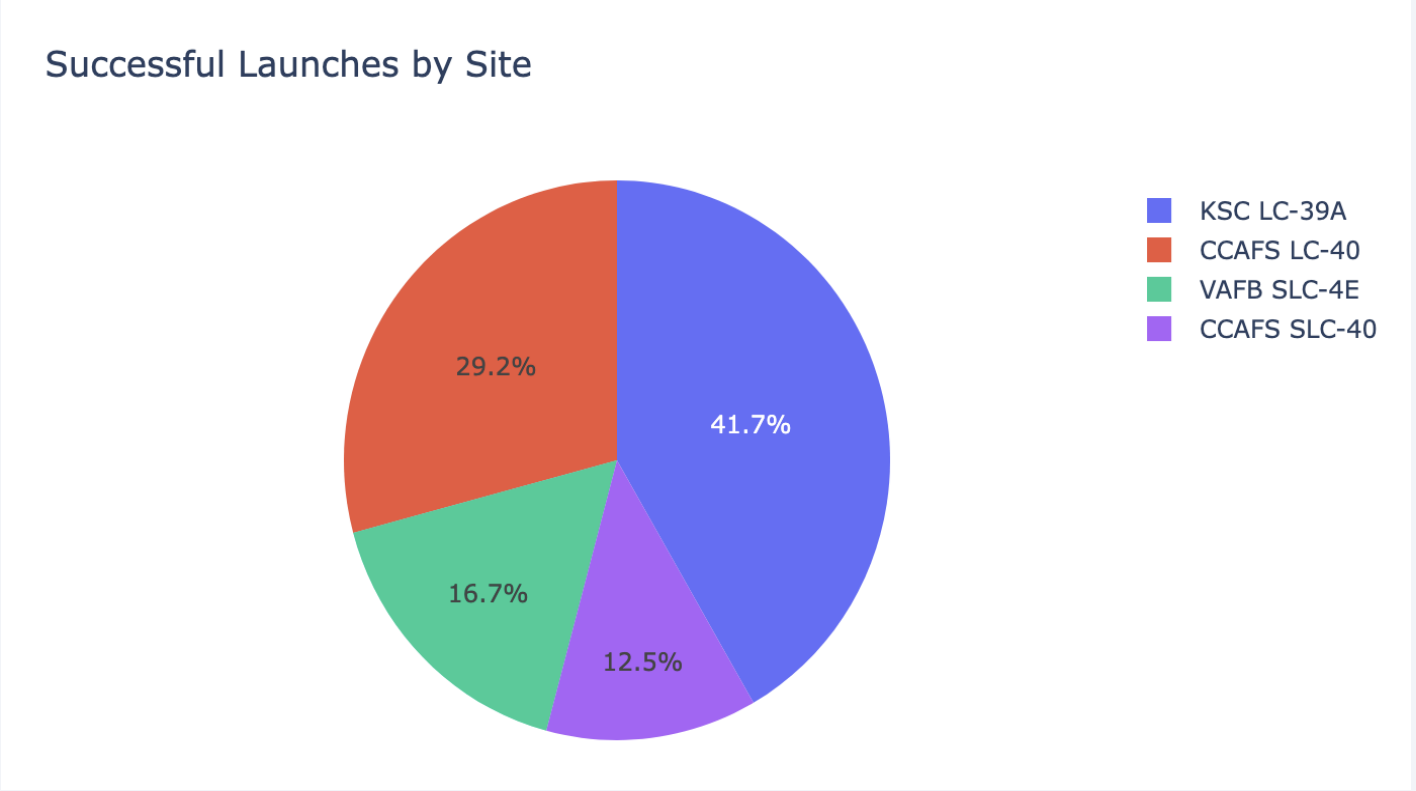
- Nearest city: 17.93KM
- Nearest rail: 21.95KM
- Nearest road: 0.59KM
- Nearest coastline: 0.86 KM




Section 4
Build a Dashboard with Plotly Dash

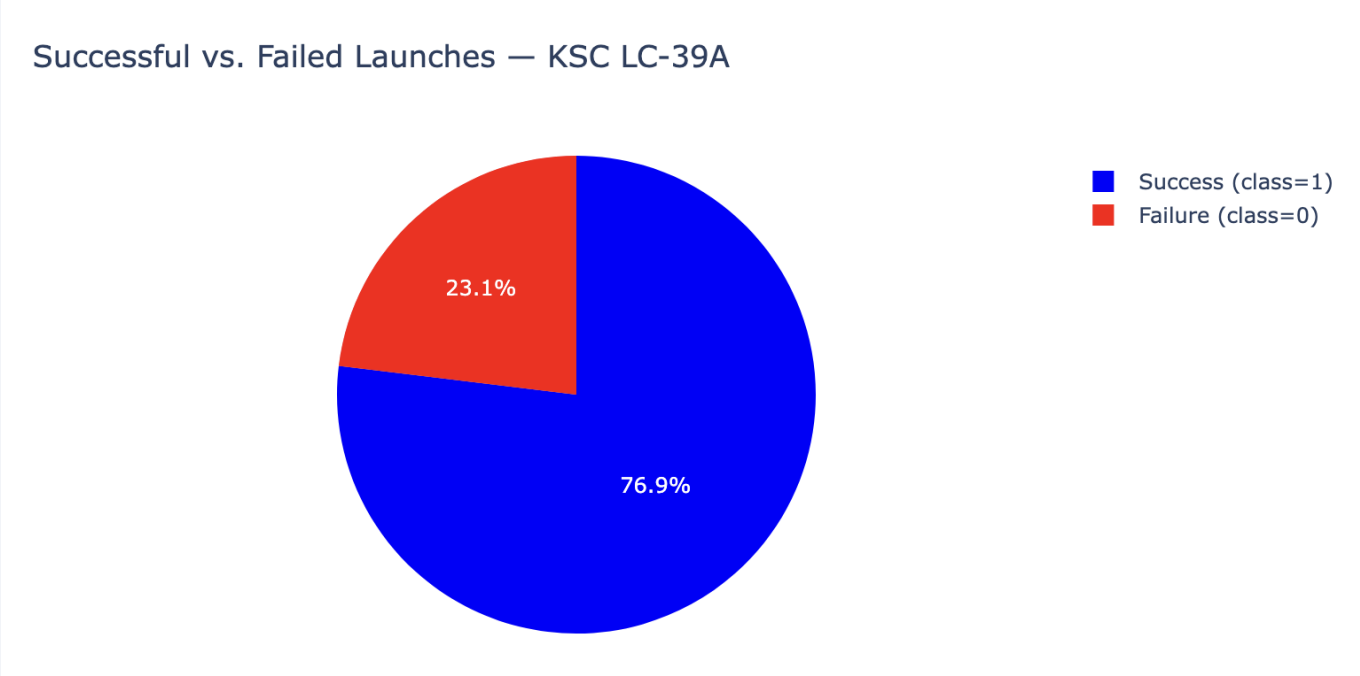
Launch Success Count for All Sites in Dash

- The chart shows a relative comparison of successful rocket booster recoveries across 4 launch sites
- KSC LC-39A is in a commanding lead with 41.7% of total recoveries coming from this site. The site had roughly 30% more successful recoveries relative to second place, CCAFS LC-40



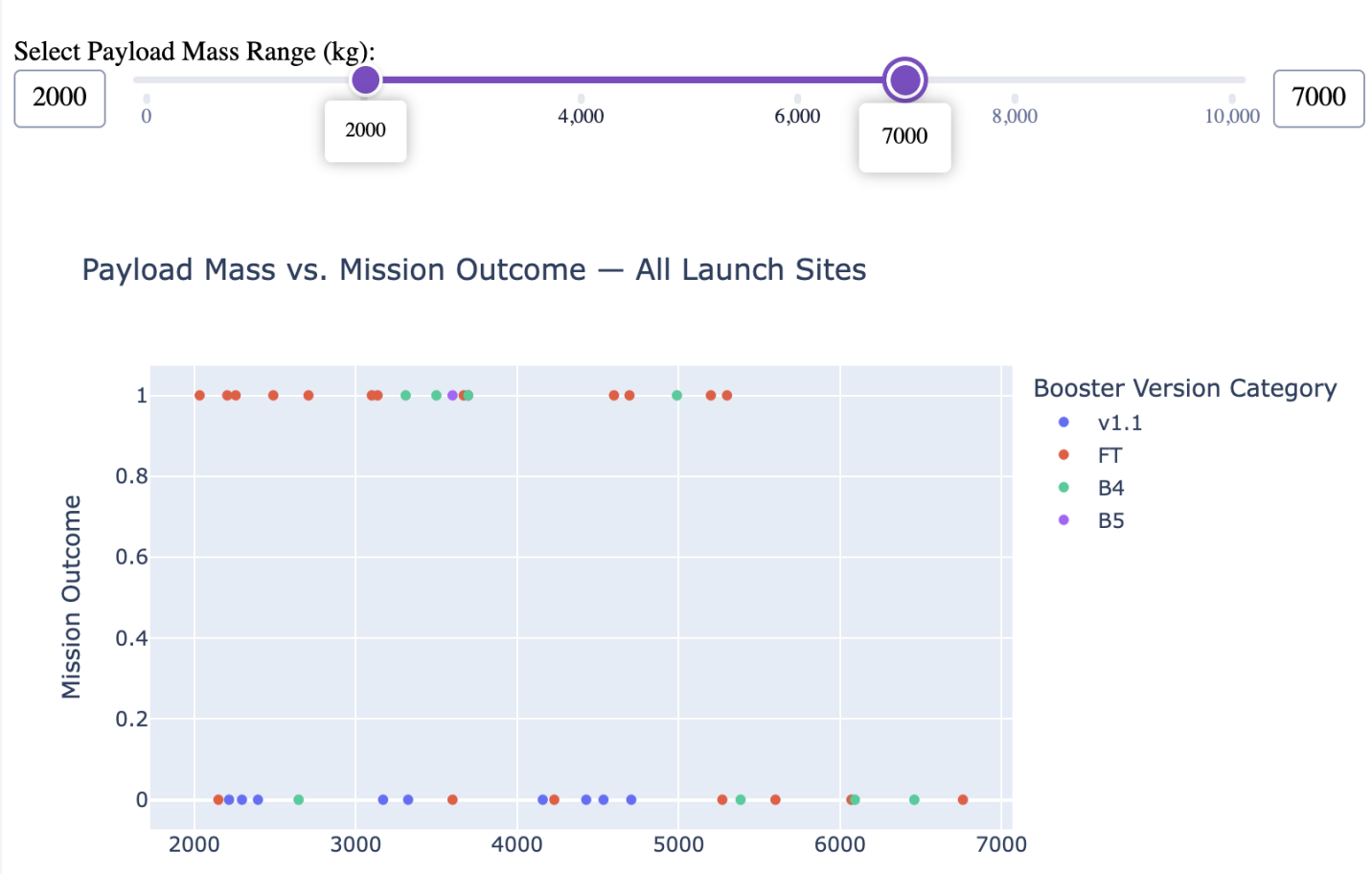
Most Successful Launch Site

- Launch site KSC LC-39A has had the most success in its recovery of Falcon 9 stage 1 rockets
- Total Successful Recovery is at 76.9% compared to just 23.1% for failed recoveries



Payload vs Launch Outcome Plot

- Narrowing the selectable Payload range to 2000-7000 kg, we see that heavier payloads have a higher failure rate especially above 5,500 kg
- Booster Version 'FT' is by far the most successful in terms of rocket recovery overall in the range of 2000 – 5,500 kg

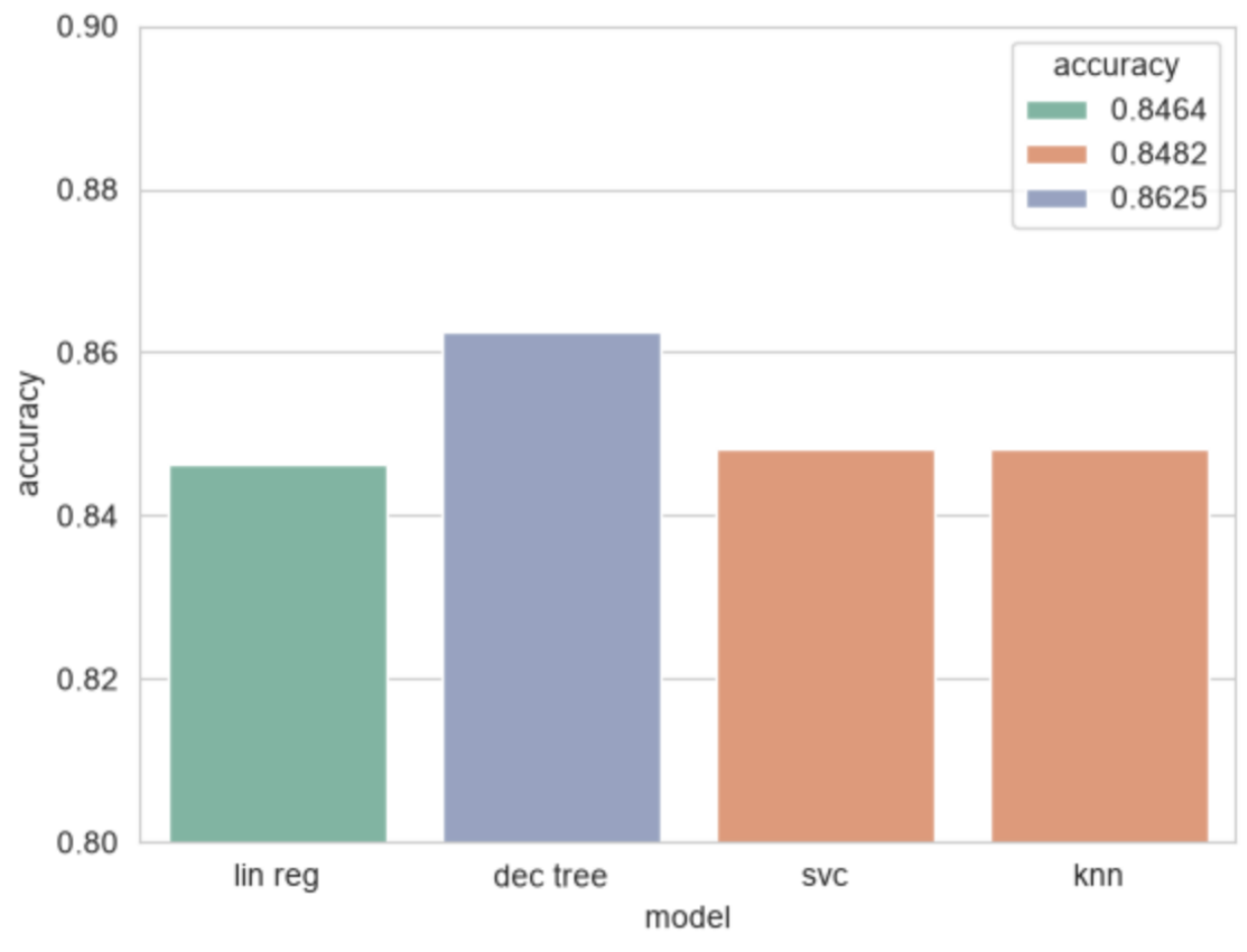


Section 5

Predictive Analysis (Classification)



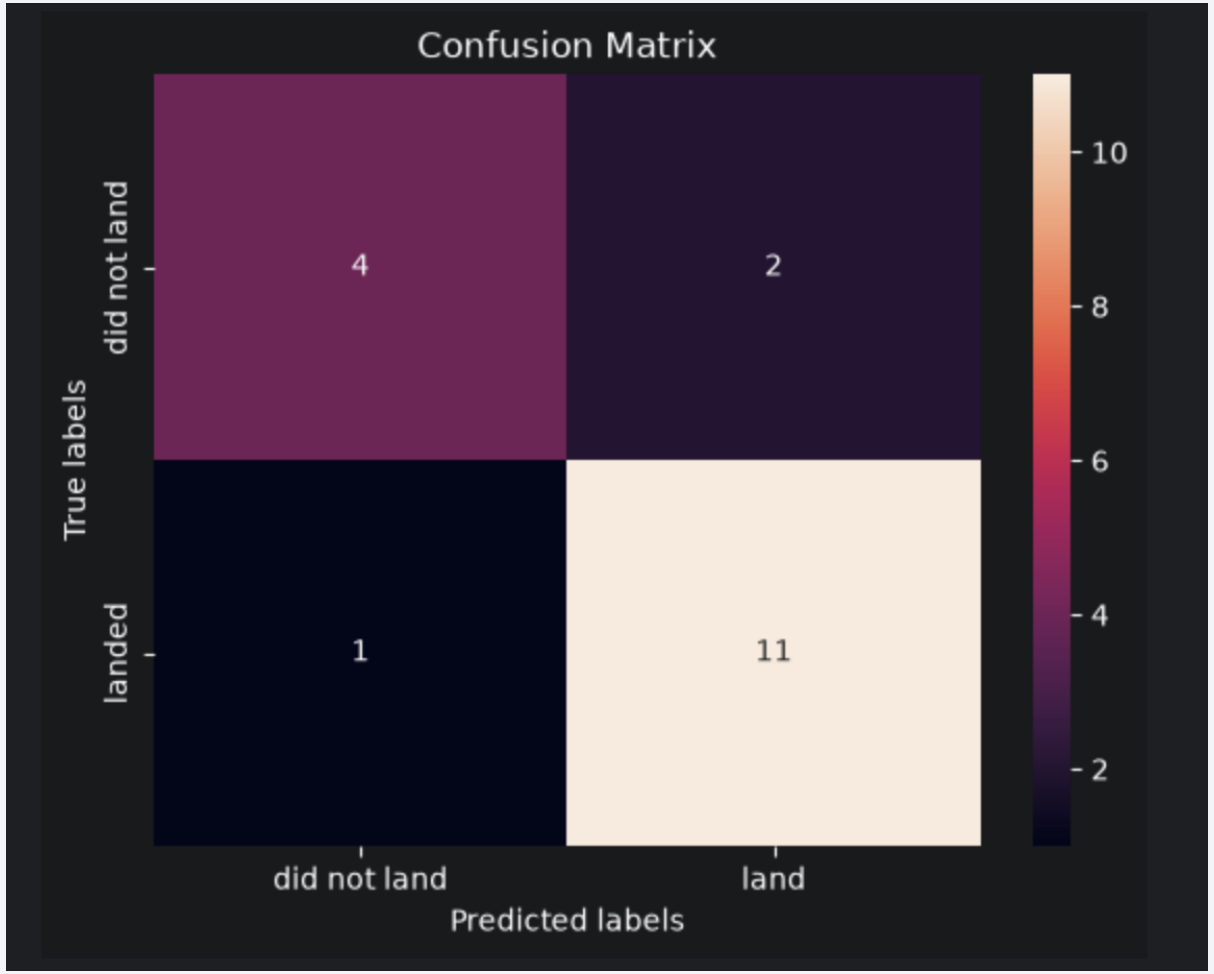
Classification Accuracy



- The Decision Tree classification model had the highest accuracy score at 86.25%

Confusion Matrix

- Presented at the right is a confusion matrix for the Decision Tree Classifier which performed the best overall out of the models built
- A confusion matrix provides a quick view into how well the model can accurately predict values. The upper left box at (4) shows correct failed predictions that indeed did fail (true negatives) and the bottom right at (11) shows successful landings that were predicted correctly (true positives)
- The model was not perfect and exhibited slightly more false positives (2) where a rocket did land but the model indicated it failed. Performance was better for false negatives with only (1) case



Conclusions 1

- The culmination of the project was to build a model that could predict the successful landing of a Falcon 9 stage1 rocket booster. By developing a model, SpaceX could have a significant competitive edge by being able to provide similar launch services at a saving of +100M per launch
- EDA uncovered that the Stage 1 Rocket Recovery Success Rate was: 66.67% across all launch sites
- CCAFS SLC 40 had the most successful launches at 55
- The highest orbit type count was ISS with 21
- The most common land type was True ASDS at 41
- More flight attempts lead to more successful recoveries. As flight tests are carried out it is clear that the teams learn from failed outcomes and improve processes as well as capture technologies
- CCAFS SLC 40 supported light and heavy payloads, with VAFB SLC 4E focused more on light payloads, KSC LC39A covered the entire range of payload mission weights
- The LEO orbit appears to have better success with more flights being executed. This is not the case for GTO that has intermittent failures through its flight number missions
- For heavier payloads the positive landing rates are higher for ISS, PO and VLEO
- The success rate since 2013 continued to increase until 2017 when there was a slight decrease into 2018. Things quickly picked back up climbing above 80% success into 2020

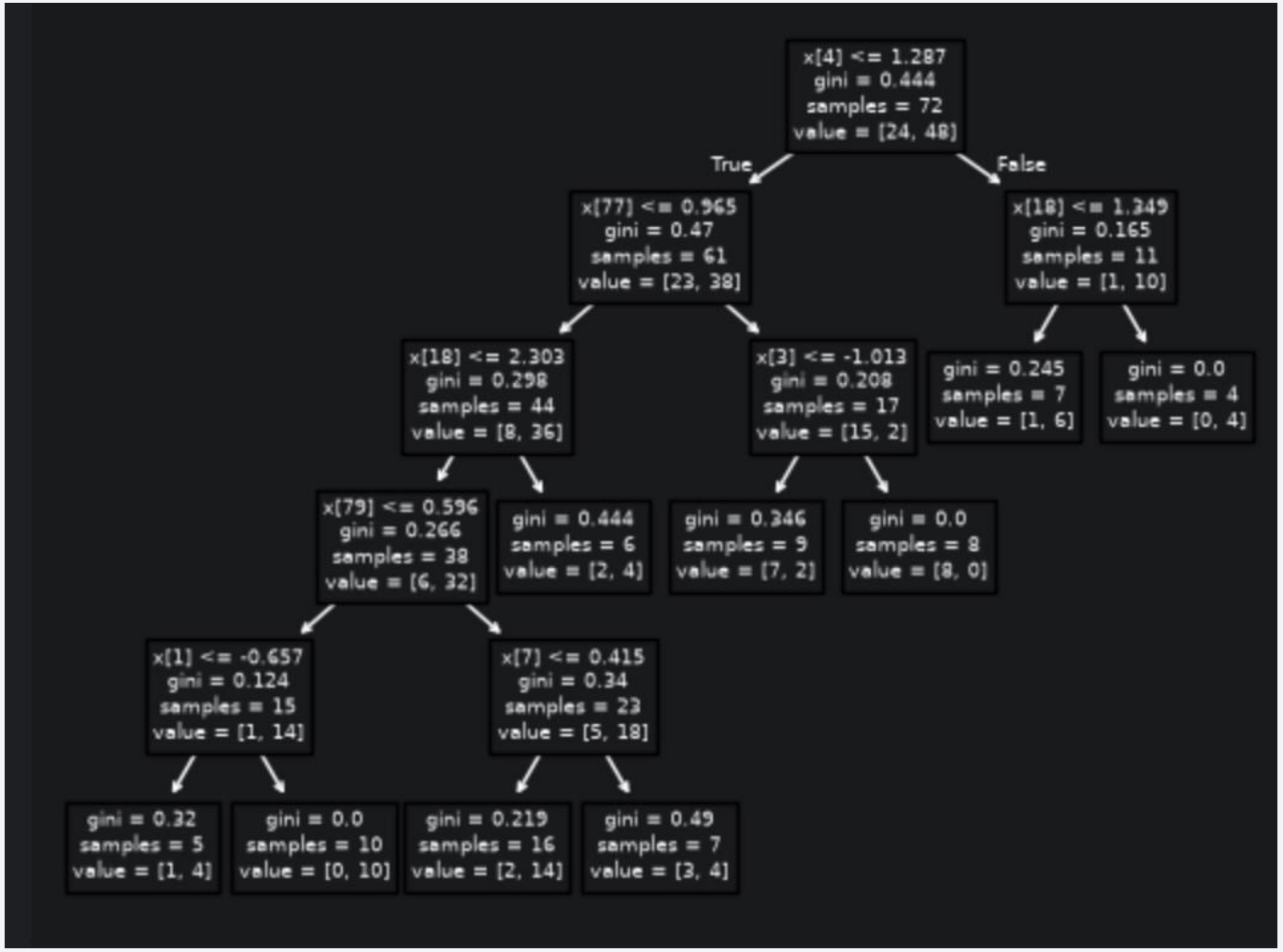
Conclusions 2

- Total Payload Mass (kg) for NASA was 45,596
- The average Payload Mass (kg) for v1.1 Falcon 9 rockets was 2,534.66
- Insights were also drawn to demonstrate the distance of key features from launch sites. Key distances for launch site CCAFS SLC 40 were: nearest city: 17.93KM, nearest rail: 21.95KM, nearest road: 0.59KM, nearest coastline: 0.86 KM
- Through exhaustive research, data collection, wrangling, EDA, machine learning model building, testing and evaluation, a DecisionTreeClassifier from scikit-learn performed the best overall with the highest accuracy at 86.25% with low false positives and false negatives. The model allowed for the most accurate prediction for successful rocket landings and could be used to net SpaceX a competitive edge and positively impact revenue growth in future business operations

Appendix

- Reference links to code snippets, visualizations, explanations, etc. can be found linked throughout the presentation via GitHub URLs
- One of my favorite coding features used throughout the project were magic commands. I deployed these heavily to display visualizations inline and execute more than one type of code within a single notebook, such as, Python and SQL. %%!

Appendix 2 – Decision Tree Map



Appendix 3 – Decision Tree ML with Unseen Data

- 10 random samples were created and run through the newly created Decision Tree model
- Based on the samples, the model graded and predicted whether those feature values would result in successful launch recoveries
- The model returned 6 failed recoveries and 4 successful recoveries
- This type of exercise is useful in being able to programmatically test the likelihood of successful returns vs failures by changing input variables and predicting outcomes vs having to actually build and fly actual rockets!!

```
yhat3 = tree_mod.predict(df_testsamp1)
✓ [158] 19ms

yhat3
✓ [159] 30ms
array([1, 1, 1, 0, 0, 0, 1, 0, 0, 0])

for item in yhat3:
    if item != 0:
        print('the rocket WAS recovered successfully')
    else:
        print('the rocket was NOT recovered successfully')
✓ [160] 21ms
the rocket WAS recovered successfully
the rocket WAS recovered successfully
the rocket WAS recovered successfully
the rocket was NOT recovered successfully
the rocket was NOT recovered successfully
the rocket was NOT recovered successfully
the rocket WAS recovered successfully
the rocket was NOT recovered successfully
the rocket was NOT recovered successfully
the rocket was NOT recovered successfully
```

Thank you!

